REPORT

EDUCATIONAL DROPOUT AND REINTEGRATION ANALYTICS DATA WAREHOUSE

Prepared By :

Aline Aloulou Mahjoub

Teacher :

MR Kamel Garrouch

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1. Introduction

1.1 Presentation of the context and subject of analysis

Educational dropout represents a major social and economic challenge in **Tunisia**. Each year, a significant number of students leave the educational system before completing their studies due to various factors such as **financial** and **family problems**, **academic difficulties**, **early marriage** and **limited access to educational infrastructure**. This phenomenon has serious long-term consequences, including increased unemployment, social exclusion, and negative impacts on national development.

In response to this issue, several reintegration programs have been implemented by governmental institutions and non-governmental organizations to provide second chances for students who have dropped out. However, evaluating the effectiveness of these programs remains difficult for decision makers due to the fragmentation of data, the heterogeneity of data sources, and the lack of structured analytical systems.

This project addresses these challenges by designing and implementing a Data Warehouse and Business Intelligence system that integrates dropout and reintegration data from multiple regions and heterogeneous sources. The proposed system enables efficient data analysis and supports informed decision-making by providing reliable indicators and insights related to educational dropout and reintegration outcomes.

1.2 Project Objectives

The main objective of this project is to design a decision support data warehouse dedicated to the analysis of educational dropout and reintegration in Tunisia.

Specific Objectives

- Integrate data from multiple sources from two Tunisian Regions (relational databases(MySQL), CSV files)
- Design a multidimensional data warehouse using a star schema
- Implement a complete ETL process using Python
- Ensure data quality via automated ETL processes through cleaning, validation, incremental loading
- Generate analytical indicators (KPIs) and visualizations for analysis
- Support decision-making for educational policies and reintegration programs

2. Detailed description of the subject of analysis

2.1 Decision Making issues

The data warehouse helps answer important questions that decision makers face at different levels:

- **Strategic Level:**
  At this level, the focus is on the big picture. It helps answer questions like:
  - ➢ Which regions or education levels need the most attention?
  - ➢ How should funding be shared between different reintegration programs?
  - ➢ Are there patterns that show which students might drop out in the future?

- **Tactical Level:**
  Here, the goal is to improve ongoing programs. The warehouse helps understand:
  - ➢ Which reintegration programs are actually working best?
  - ➢ Which regions would benefit from new programs?
  - ➢ Are there specific times in the year when dropout rates are higher?

- **Operational Level:**
  At the daily level, the warehouse helps schools and programs support students directly:
  - ➢ Which students are at the highest risk of dropping out?
  - ➢ Which programs fit each student's needs the most?
  - ➢ How fast are students getting reintegrated into school?

2.2 Stakeholders Involved

Different people and organizations are involved in dropout prevention and reintegration, and each has its own needs:

Stakeholder	Role	Decision Needs
Ministry of Education	Policy makers	National trends, which regions need priority, how to distribute budgets
Regional Education Offices	Program managers	Regional dropout patterns, where to focus programs
Reintegration Centers	Service providers	Which programs work best for different students
Educational Researchers	Analysts	Understanding dropout causes and program impacts
School Administrators	Implementers	Identifying students at risk and monitoring reintegration progress

2.3 Key Questions Answered by the BI System

Dropout analysis

1. What is the overall dropout and reintegration rate? (KPI 5)
2. Which regions (Tunis, Sousse) have the highest dropout rates?(KPI 1)
3. At which education levels (Primary, Middle, High School, Higher Education) do most dropouts occur?(KPI 6)
4. What are the primary reasons for educational dropout?(KPI 11)
5. How do dropout rates vary by gender and age group?(KPI 9 , 8 and 7)
6. What are the dropout trends over time?(KPI 3)

7. How has the situation evolved year by year?(KPI 10)

Reintegration Program Effectiveness

8. Which reintegration programs are most successful?(KPI 4)
9. How long does successful reintegration typically take?(KPI 2)

Regional Analysis

13. Which regions have the highest dropout rates? (KPI 11)
14. How do reintegration success rates vary by region? (KPI 11)

Education System & Level Analysis

17. Which education levels experience the highest dropout rates? (KPI 6)
18. Are students at certain education levels more likely to successfully reintegrate? (KPI 6)

Risk Factor Analysis

19. Which age groups are most at risk of dropping out? (KPI 7)
20. How does dropout affect male vs female students differently?(KPI 9)
21. What are the primary reasons students drop out, and do they vary by gender? (KPI8 and KPI 11)

3. Data Warehouse Design

Designing a data warehouse is one of the most important steps in a BI project because it determines how easy it will be to store, access, and analyze data. For this project, I wanted to create a system that allows decision-makers to quickly get insights about educational dropout and reintegration in Tunisia.

3.1 Choice of Schema Type

When we design a data warehouse, one of the first decisions is which schema to use. A **schema** is basically the blueprint of how data will be organized in the warehouse. There are different types of schemas, like **star**, **snowflake**, and **constellation**, but I chose the **star schema** for this project.

Why I chose the star schema

- **Simplicity:** The star schema is very simple. It has one central table (the fact table) and several tables connected to it (dimension tables). This makes it easy to understand, even for people who are not very technical. I thought this was important because school administrators or program managers might use this data.

- **Query performance:** With the star schema, it's easier and faster to run queries. Since all dimensions are directly linked to the fact table, we don't need many joins. This is useful because some reports, like dropout trends over several years, could involve lots of data.

- **Clear organization:** Each dimension table has information about a specific category, like students, regions, or programs. The fact table contains the measurements we want to analyze, like the number of dropouts or the reintegration status. This separation makes the warehouse neat and logical.

- **Compatibility with BI tools:** Star schema works very well with visualization tools like Power BI or Python dashboards. I wanted to make sure the warehouse could be used easily for creating charts, dashboards, and KPIs.

Other schema options I considered

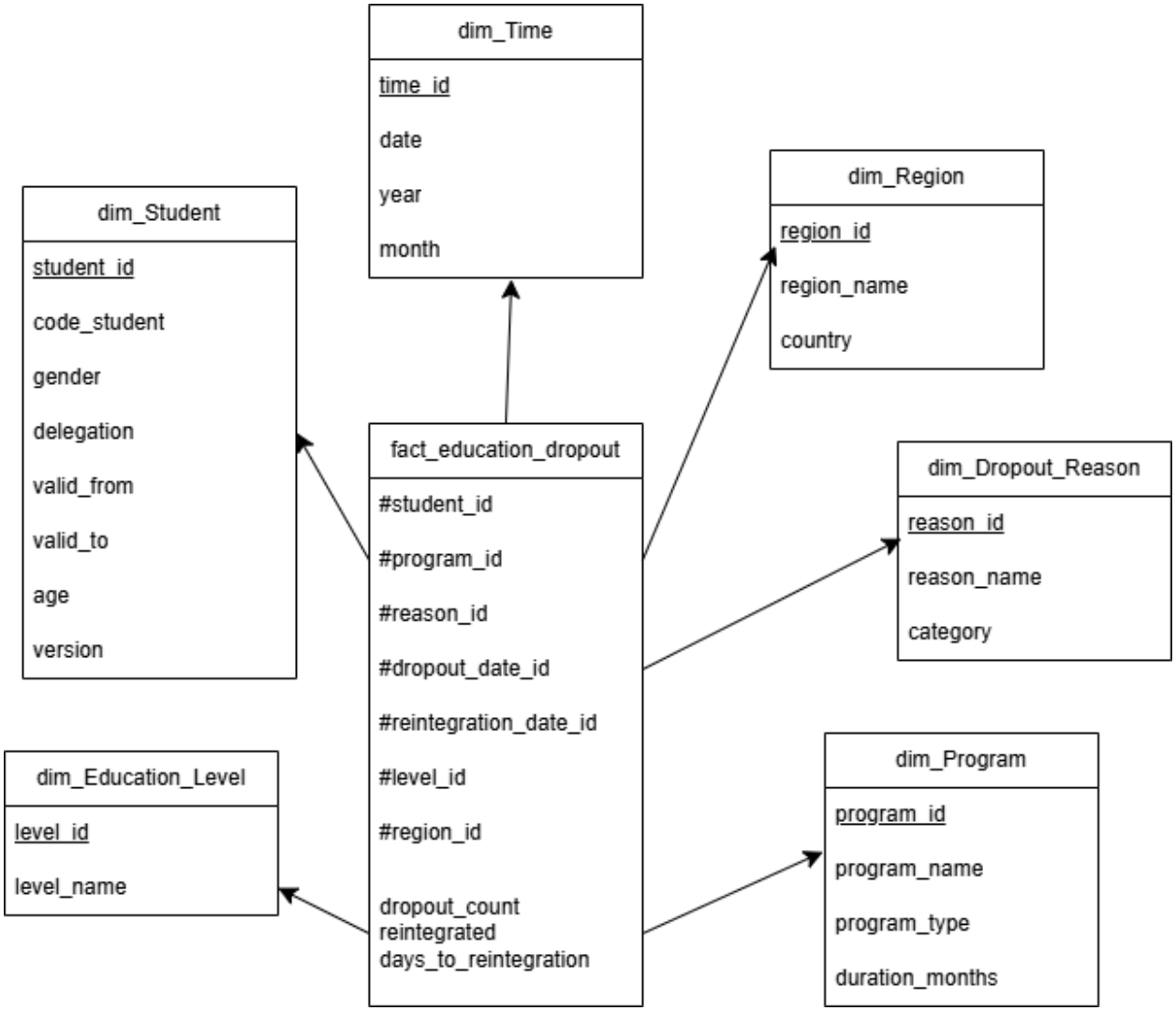
- **Snowflake schema:** I thought about normalizing the dimension tables to save space, like in a snowflake schema. But I realized it would make queries more complex because we would need more joins. For this project, clarity and speed were more important than saving storage.

- **Constellation schema:** This is used when multiple fact tables share dimension tables. I didn't choose this because my analysis focuses on one main fact: student dropout and reintegration. There is no need for multiple fact tables in this case.

So, after thinking about simplicity, speed, and usability, I decided that the **star schema** is the best option for this project.

3.2 Multidimensional Model

After choosing the star schema, I designed the data warehouse like this:



Fact Table

fact_education_dropout: This table is the center of the star. It contains measurable information about each dropout event, The attributes I included are:

- **student_id**: Links to dim_student
- **region_id**: Links to dim_region

- **education_level_id**: Links to dim_education_level
- **program_id**: Links to dim_program
- **reason_id**: Links to dim_dropout_reason
- **dropout_date_id**: Links to dim_time
- **reintegration_date_id**: Links to dim_time
- **reintegrated**: Boolean value to show if the student returned
- **days_to_reintegration**: Number of days until reintegration
- **dropout_count**: Counts dropouts

Dimension Tables

Dimension tables are the backbone of a star schema because they **describe the facts** in the fact table. Each dimension adds context and makes it possible to answer specific business questions. In this project, I designed six dimensions: **Student, Time, Region, Education Level, Program, and Dropout Reason**.

1. **dim_student:**

   This table contains information about students and their personal characteristics. It allows us to analyze dropout patterns by gender, age...

➢ I used **Slowly Changing Dimension Type 2 (SCD Type 2)**. This means that if a student's information changes over time (like age or delegation), we keep a **history** of the changes. This is important because we want to analyze dropout events using the **student's status at the time of the event**, not the current status

2. **dim_time:**

   Contains details about dates so that we can analyze dropouts by year, month, or day.

3. **dim_region:**

   Stores information about geographic locations(Tunis , Sousse) to compare dropout and reintegration rates across areas.

4. **dim_education_level:**

   Tracks the student's current level (primary, secondary, higher education) to see which levels are most affected.

5. **dim_program:**

   Contains details about reintegration programs offered to students who dropped out. This helps **analyze program effectiveness**.

6. **dim_dropout_reason:**

   Explains the cause of dropout (financial, family, academic, etc.) so that we can understand trends and target interventions.

4. Data Sources

For this project, I used data from three different sources in Tunisia: Tunis, Sousse, and Kairouan. Each source has a different format and structure, which helps the data warehouse handle diverse types of data.

- Tunis (Source A): MySQL database with structured tables (~150 records)
- Sousse (Source B): CSV files exported quarterly (~85 records)

In total, I collected about 235 dropout records covering 2021–2024.

4.1 Description of Sources Used

4.1.1 Tunis Relational Database

The Tunis data comes from an operational MySQL database. It has structured tables that keep track of students, their dropout information, and if they were reintegrated into any program.

The main tables are:

- **students:** Basic info like name, birth date, gender, school, and enrollment details.
- **dropout_records:** When the student dropped out, why, and some school info.

- **reintegration_tracking:** Tracks programs students joined, whether they succeeded, and the cost of the program.

Characteristics:

- Maintains referential integrity with foreign keys.
- Contains detailed student information: names, date of birth, gender, enrollment, school, and education level.
- Dropout records include reasons, school distance, and family income.
- Reintegration tracking provides program details, success status, start and completion dates, and program cost.

Challenges:

- Dropout reasons and program names need to be standardized.
- We have to calculate age from the birthdate.
- Combining information requires joining multiple tables.

4.1.2 Sousse CSV Files

Sousse data consists of **quarterly CSV files**, covering 2021–2024.

Structure: Columns are in French and include student ID, full name, date of birth, dropout date, education level, reason for dropout, delegation, age, gender, reintegration status, program, school distance, family income, and internet access.

Data Quality Issues:

- Duplicate student IDs
- Missing values for key fields like student ID, dates, or income.
- Multiple date formats (YYYY-MM-DD, DD-MM-YYYY, MM/DD/YYYY).
- Not Standardized school level or program names.

- Boolean values and gender labels vary (“Oui”, “Yes”, “1” for TRUE; “M”, “Masculin” for male).

ETL Transformations Required:

- Deduplication and missing value handling.
- Standardization of dates, text, and boolean fields.
- Gender Translation
- Converting age and ids column into numeric values
- Adding new columns like region_id , program_id , days_to_reintegration ect

After cleaning, the Sousse data became consistent, with correct values, no duplicates, and standardized formats.

4.2 Limitations of the Data

Even after cleaning, the data still has some limitations:

- **Temporal Limitations :** The data only covers 2021–2024. We don’t have records before 2021, so we cannot analyze trends beyond these four years
- **Geographic Limitations:** Only 2 of 24 governorates included. Some regions, like Sfax and Bizerte, have no data.
- **Completeness Limitations:** Not all socioeconomic details are available. Information about schools’ facilities, program costs, or resources is limited. There is also no follow-up data after reintegration to see long-term outcomes.
- **Quality Differences:** Some sources, like Sousse CSV, had many errors and missing values. Different coding systems across the 2 sources made it challenging to combine them.

5. ETL Process

The ETL (Extract, Transform, Load) process is a crucial step in building our education dropout data warehouse. It ensures that raw data from multiple sources is cleaned, transformed, and loaded into a structured format that can be easily analyzed. In this project, we implemented the ETL process using **Python**, **Pygrametl**, **PostgreSQL**, and **MySQL**.

5.1 Extraction

I used two main sources for extraction:

1. **CSV Files (Sousse Region)**
   - Data for the Sousse region was provided as a CSV file named **cleaned_sousse_dropout.csv**.
   - This file contains information about students who dropped out, including gender, age, education level, dropout reason, program type, and reintegration data.
   - The CSV was read in Python using **Pygrametl's CSVSource** object, which allows row by row processing for efficient handling of large files.
2. **Relational Database (Tunis Region)**
   - The Tunis region data was stored in a MySQL relational database named **tunis_education_system.**
   - Tables included students, dropout_records, and reintegration_tracking.

- Data was extracted using SQL queries wrapped in a **Pygrametl SQLSource**, which enabled direct mapping of database rows into our warehouse tables.

During extraction, we applied **incremental loading** by checking the last successful ETL run for each source. This prevents duplicate records and ensures only new or updated data is processed.

5.2 Transformation

The transformation step involved several operations:

1. **Cleaning and Standardization**
   - Dates were converted into a consistent format (**YYYY-MM-DD**) for both dropout and reintegration events.
   - Missing values in fields such as age, gender, and reintegration days were handled by either computing the missing values (calculating days between dropout and reintegration) or setting default values.
   - Categorical fields like dropout reason, program type, and education level were mapped to standardized values to match our dimension tables.
2. **Dimensional Mapping**
   - Each student, program, dropout reason, and region was matched to its respective dimension table using lookup attributes.
   - The **slowly changing dimension (SCD)** technique was applied for the student dimension to track historical changes, like updates to gender or age, while maintaining previous versions.

3. **Fact Table Preparation**
    - For each record, a fact row was created linking the student, dropout reason, program, region, and time dimensions.
    - Measures such as reintegrated , days_to_reintegration, and dropout_count were calculated.
    - Repeated students were tracked to increment the **dropout_count**.

5.2 Load

Data was loaded into the warehouse using Pygrametl dimension and fact objects:

- **Dimensions**
    - **dim_student (SCD Type 2):** Tracks historical changes in student attributes.
    - **dim_education_level:** Stores education levels such as Primary, Middle School, High School, and Higher Education.
    - **dim_program:** Contains program details including name, type, and duration.
    - **dim_dropout_reason:** Holds reasons for dropout with categories.
    - **dim_region:** Stores the names and countries of regions.
    - **dim_time:** Stores all dates from 2021-01-01 to 2025-12-31 with day, month, and year breakdown.

- **Fact Table**
    - **fact_education_dropout** stores facts about each dropout event, linking all dimensions and storing measures like reintegration status and days to reintegration.

Static dimension data (regions, programs, education levels, and dropout reasons) were preloaded. Time dimension was fully populated with daily records for the 5 year range to simplify reporting.

5.4 ETL Control and Monitoring

To ensure ETL reliability, a control table named **etl_control** was created in PostgreSQL. This table tracks:

- Source name (CSV or database)
- Last successful ETL run date
- Number of records processed
- Status of the ETL

Before loading data, the ETL process checks the last success date to only process new data. After loading, the control table is updated to record the new successful load and the number of records processed.

5.5 ETL Statistics and Data Quality Report

To ensure transparency and demonstrate the effectiveness of the ETL process, detailed statistics were tracked for each data source.

6. Implementation

In this project, the ETL process and data visualization were implemented using a combination of **Python** and **SQL**, creating a complete pipeline from raw data extraction to meaningful analysis.

6.1 Technology Stack

To build the project, I used the following technologies:

- **Python**: The main programming language for writing the ETL scripts and generating visualizations. Python was chosen for its simplicity and powerful libraries for data manipulation and analysis.
- **Pygrametl**: A Python library specifically designed for ETL workflows, which helped manage dimensions, fact tables, and slowly changing dimensions efficiently.
- **PostgreSQL**: Used as the data warehouse to store processed and transformed data. It also hosts the control table for ETL monitoring.
- **MySQL**: Source database for Tunis region data. Python connected to it to extract student, dropout, and reintegration information.
- **Pandas**: Used to handle CSV files, clean data, and transform it before loading into the warehouse.
- **Dateutil**: To parse and manage date columns in different formats.
- **Matplotlib**: A Python library used to create static, customizable visualizations such as bar charts, line charts, and histograms.

- **Seaborn**: Built on top of Matplotlib, Seaborn was used for creating more advanced and aesthetically pleasing statistical visualizations.

- **NumPy**: Used for numerical operations and array manipulations, especially when preparing data for visualization.
- **SQL**: Used both in MySQL for extracting source data and in PostgreSQL for control table management and verifying data in the warehouse.

6.2 SQL Scripts Used For The Data warehouse

- **Dimensions:**

```sql
CREATE TABLE dim_student (
    student_key SERIAL PRIMARY KEY,
    student_id INTEGER NOT NULL,
    gender VARCHAR(10),
    age INTEGER,
    delegation VARCHAR(100),
    valid_from DATE NOT NULL,
    valid_to DATE,
    is_current BOOLEAN DEFAULT TRUE
);
CREATE TABLE dim_time (
    time_id SERIAL PRIMARY KEY,
    date DATE UNIQUE NOT NULL,
    day INTEGER,
    month INTEGER,
    year INTEGER,
    quarter INTEGER,
    semester INTEGER
);

CREATE TABLE dim_region (
    region_id SERIAL PRIMARY KEY,
    region_name VARCHAR(100) UNIQUE NOT NULL,
    country VARCHAR(100)
);
```

```sql
CREATE TABLE dim_education_level (
    level_id SERIAL PRIMARY KEY,
    level_name VARCHAR(50) UNIQUE NOT NULL,
    category VARCHAR(50)
);
CREATE TABLE dim_program (
    program_id SERIAL PRIMARY KEY,
    program_name VARCHAR(100) UNIQUE NOT NULL,
    program_type VARCHAR(50),
    duration_months INTEGER
);
CREATE TABLE dim_dropout_reason (
    reason_id SERIAL PRIMARY KEY,
    reason_name VARCHAR(100) UNIQUE NOT NULL,
    category VARCHAR(50),
    description TEXT
`
```

- **Fact Table:**

```sql
CREATE TABLE fact_education_dropout (
    fact_id SERIAL PRIMARY KEY,
    student_key INTEGER REFERENCES dim_student(student_key),
    region_id INTEGER REFERENCES dim_region(region_id),
    level_id INTEGER REFERENCES dim_education_level(level_id),
    program_id INTEGER REFERENCES dim_program(program_id),
    reason_id INTEGER REFERENCES dim_dropout_reason(reason_id),
    dropout_date_id INTEGER REFERENCES dim_time(time_id),
    reintegration_date_id INTEGER REFERENCES dim_time(time_id),

    reintegrated BOOLEAN DEFAULT FALSE,
    days_to_reintegration INTEGER,
    dropout_count INTEGER DEFAULT 1);
```

- **ETL CONTROL TABLE:**

```sql
CREATE TABLE etl_control
(
    control_id SERIAL PRIMARY KEY,
    source_name  VARCHAR(100) NOT NULL,
    last_success_date timestamp ,
    records_processed integer,
    status VARCHAR(20)
)
```

7. Restitution and Analysis

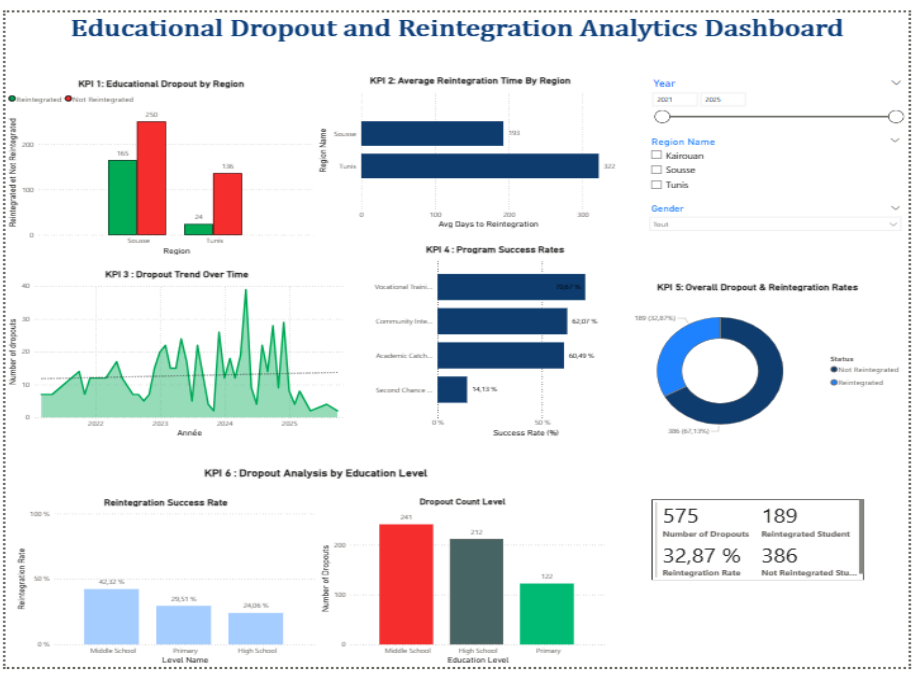
After completing the ETL process and successfully loading all the data into the data warehouse, the next step was to analyze the data and present it in a clear and understandable way. The main objective of this phase was to transform the stored data into meaningful insights that can help understand the causes of school dropout and evaluate the effectiveness of reintegration programs.

To achieve this, I developed **11 Key Performance Indicators (KPIs)**. These KPIs answer the main business questions related to dropout trends, regional differences, gender impact, education levels, and reintegration performance.

7.1 Dashboard Implementation

To visualize the KPIs, I used **two complementary tools: Python and Power BI**.

<u>Power BI Dashboard</u>



Power BI was used to create an **interactive dashboard** that allows dynamic exploration of the data. The dashboard includes:

- Filters by **year**, **region**, and **gender**
- Interactive charts that update automatically based on user selections
- A clear visual layout combining multiple KPIs on a single screen

The Power BI dashboard presents:

- Dropout and reintegration by region
- Average reintegration time
- Dropout trends over time
- Program success rates
- Education level analysis
- Global statistics (total dropouts, reintegration rate)

This interactivity makes Power BI especially useful for decision makers, as it allows them to quickly focus on specific regions or groups without writing queries.

Python Visualizations

In addition to Power BI, I also used **Python (Matplotlib and Seaborn)** to generate charts for documentation and reporting purposes.

The data was extracted directly from the **PostgreSQL data warehouse** using SQL queries executed in Python.

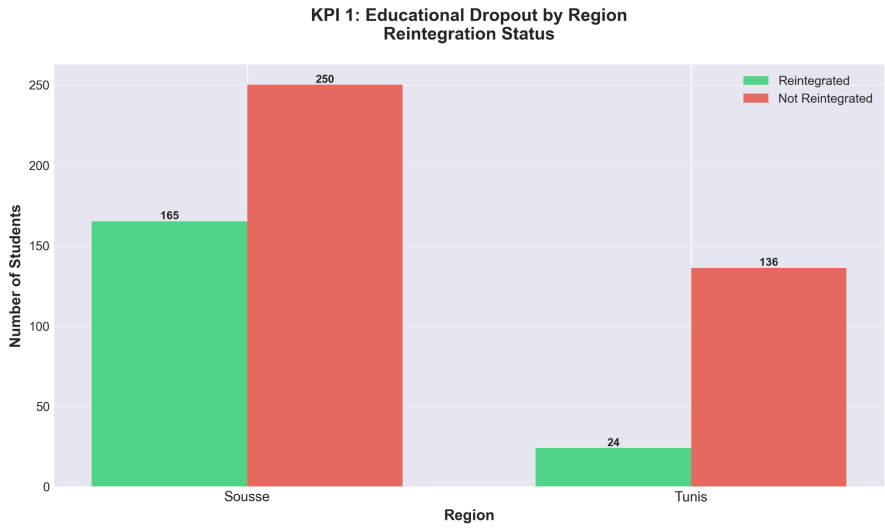
Python was mainly used for:

- Producing precise KPI charts for the written report
- Validating results shown in Power BI
- Ensuring reproducibility of analyses

7.2 KPI Analysis

The following KPIs were analyzed using both **Power BI dashboards** and **Python visualizations**, ensuring consistency and accuracy of results.

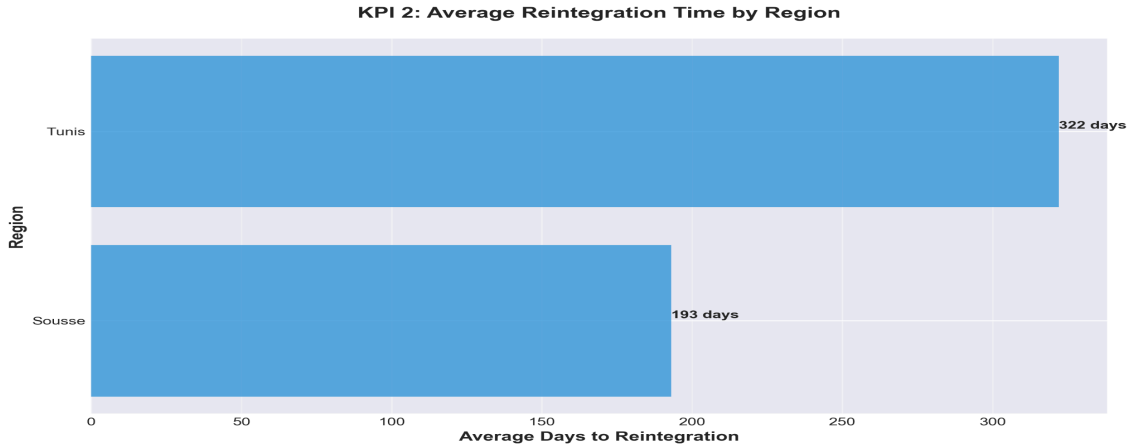
KPI 1: Educational Dropout by Region



The analysis shows clear differences between the two regions:

- Sousse has 415 total dropouts: 165 reintegrated and 250 not reintegrated.
- Tunis has 160 total dropouts: 24 reintegrated and 136 not reintegrated.
- While Sousse has 2.6 times more dropouts than Tunis, it achieves 6.9 times more reintegrations, demonstrating significantly better intervention effectiveness.

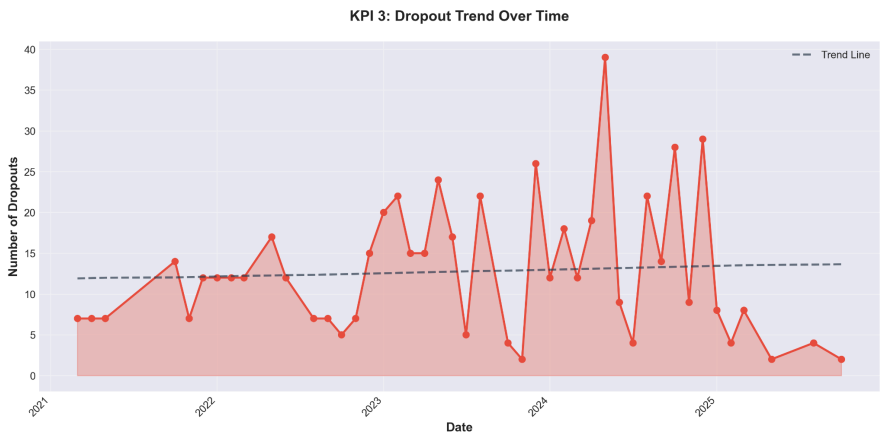
KPI 2: Average Reintegration Time by Region



This KPI measures how long students take to reenter the education system after dropping out.

- In Tunis, the average reintegration time is 322 days for the 24 reintegrated students.
- In Sousse, reintegration takes an average of 193 days for 99 students.
- This means that reintegration in Tunis takes about 67% longer than in Sousse

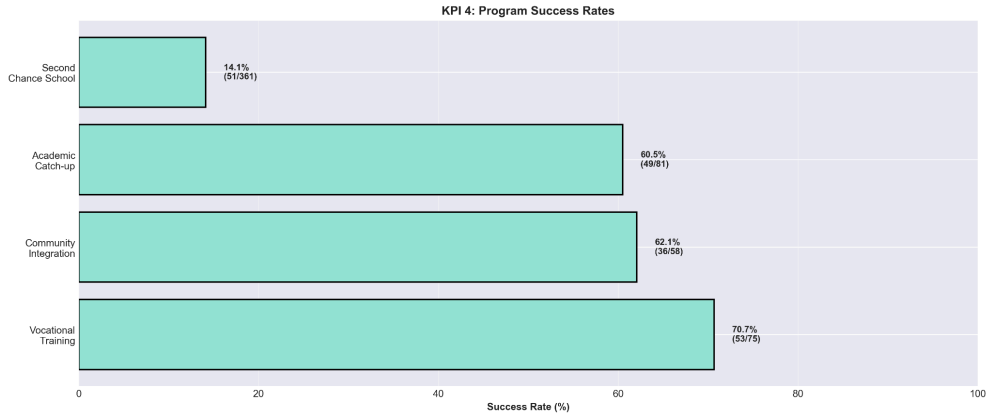
KPI 3: Dropout Trend Over Time (2021–2024)



This KPI tracks the monthly fluctuation of dropout cases from 2021 to 2025.

- Dropouts show significant volatility, ranging from as low as 2 students to a peak of 39 students per month.
- A concerning spike occurred in mid-2024 with 39 dropouts.
- The trend line remains relatively stable around 13 dropouts per month throughout the period.
- Recent data (late 2024 to early 2025) shows a dramatic improvement with dropout numbers falling to near zero.

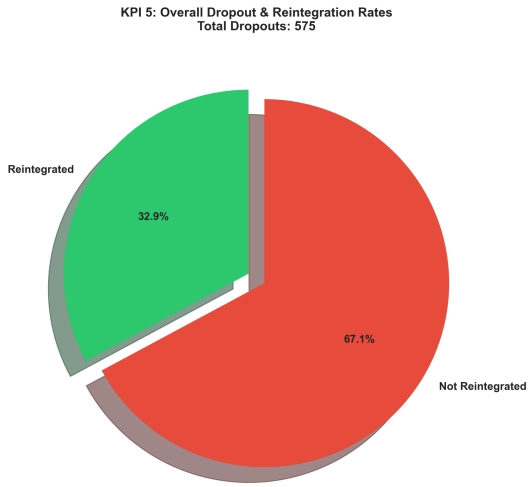
KPI 4: Program Success Rates



This KPI evaluates the effectiveness of different reintegration programs.

- Vocational Training leads with 70.7% success rate (53 out of 75 students).
- Community Integration achieves 62.1% success (36 out of 58 students).
- Academic Catch-up shows 60.5% success (49 out of 81 students).
- Second Chance School has the lowest performance at only 14.1% (51 out of 361 students).
- Vocational Training is 5 times more effective than Second Chance School, suggesting hands on skill development resonates better with at-risk students.

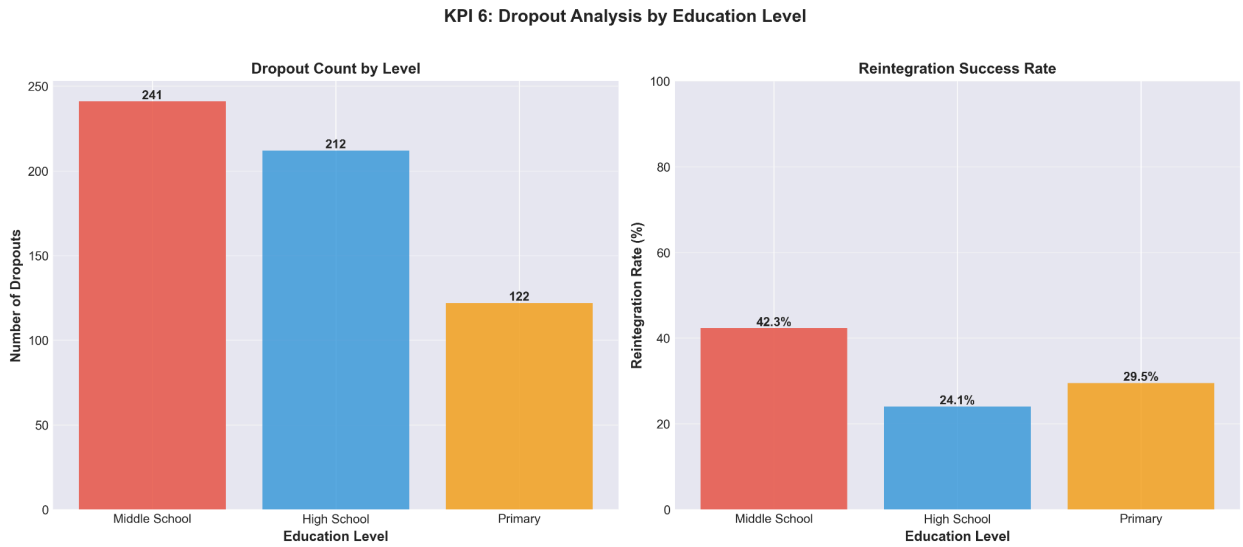
KPI 5: Overall Dropout and Reintegration Rates



This KPI provides a comprehensive view of the total dropout situation.

- Out of 575 total dropouts, only 189 students (32.9%) have been reintegrated.
- The majority, 386 students (67.1%), remain outside the education system.
- This indicates that roughly 2 out of every 3 dropout students are not successfully brought back into education, highlighting a critical gap in reintegration capacity.

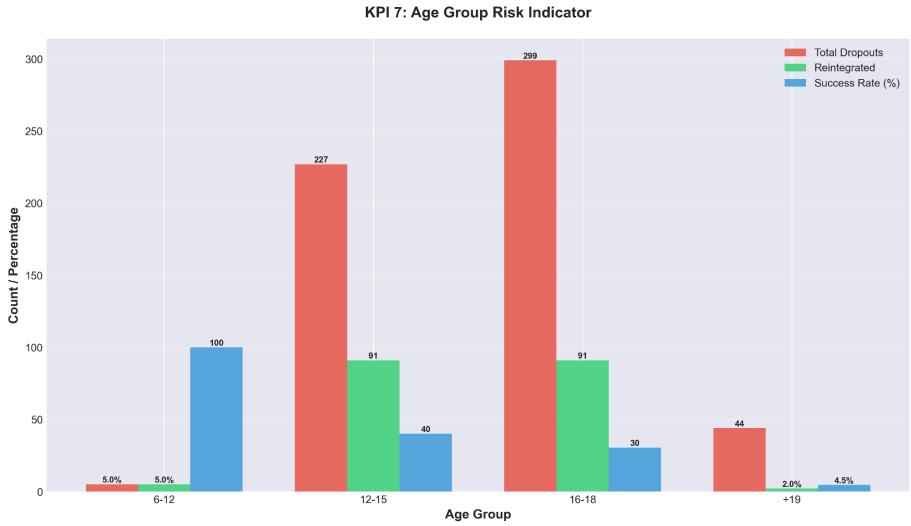
KPI 6: Dropout Analysis by Education Level



This KPI examines dropout patterns and reintegration success across different school levels.

- Middle School has the highest dropout count with 241 students, but only 42.3% reintegration rate (102 students).
- High School accounts for 212 dropouts with the poorest reintegration rate at 24.1% (51 students).
- Primary level has 122 dropouts but shows the best relative success at 29.5% reintegration (36 students).
- Despite having the most dropouts, Middle School demonstrates better recovery outcomes than High School, suggesting interventions work better for younger adolescents.

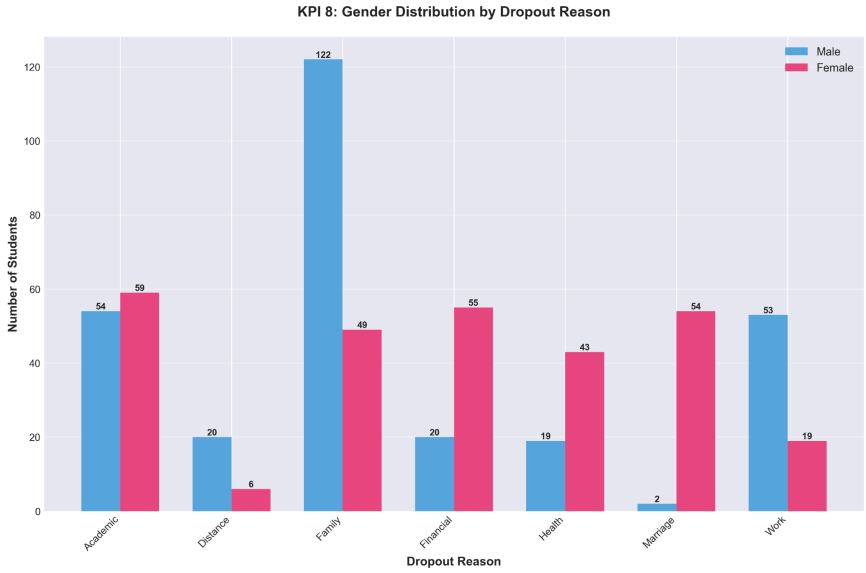
KPI 7: Age Group Risk Indicator



This KPI identifies which age groups face the highest dropout risk and recovery challenges.

- The 16-18 age group shows the highest vulnerability with 299 total dropouts (52% of all cases).
- Only 91 students from this group (30%) were reintegrated, showing the difficulty of engaging older adolescents.
- The 12-15 age group has 227 dropouts with 91 reintegrations (40% success rate).
- The 6-12 age group represents only 5% of dropouts but achieves 100% reintegration success.
- Students aged 19+ account for 44 dropouts with minimal reintegration (4.5%), indicating that once students reach adulthood, they rarely return.

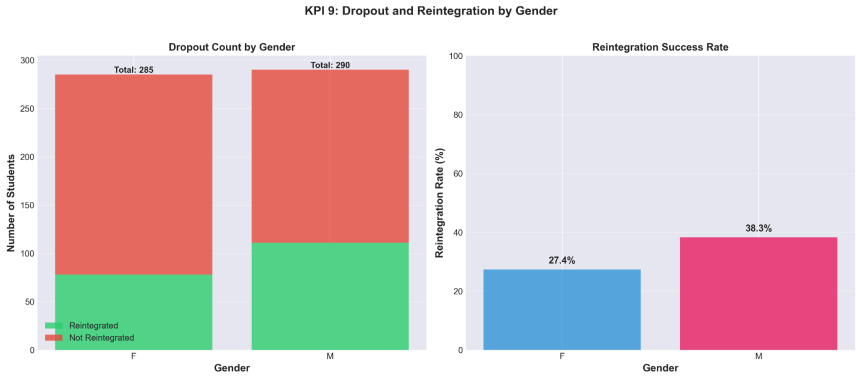
KPI 8: Gender Distribution by Dropout Reason



This KPI reveals how dropout reasons differ between male and female students.

- Family issues are the leading cause for both genders: 122 males and 49 females (171 total).
- Academic struggles affect 54 males and 59 females, showing females face slightly more academic challenges.
- Work-related dropouts are heavily male-dominated: 53 males versus only 19 females.
- Marriage affects predominantly females: 54 female dropouts compared to just 2 males.
- Financial constraints impact 20 males and 55 females, indicating economic pressures affect females more severely.

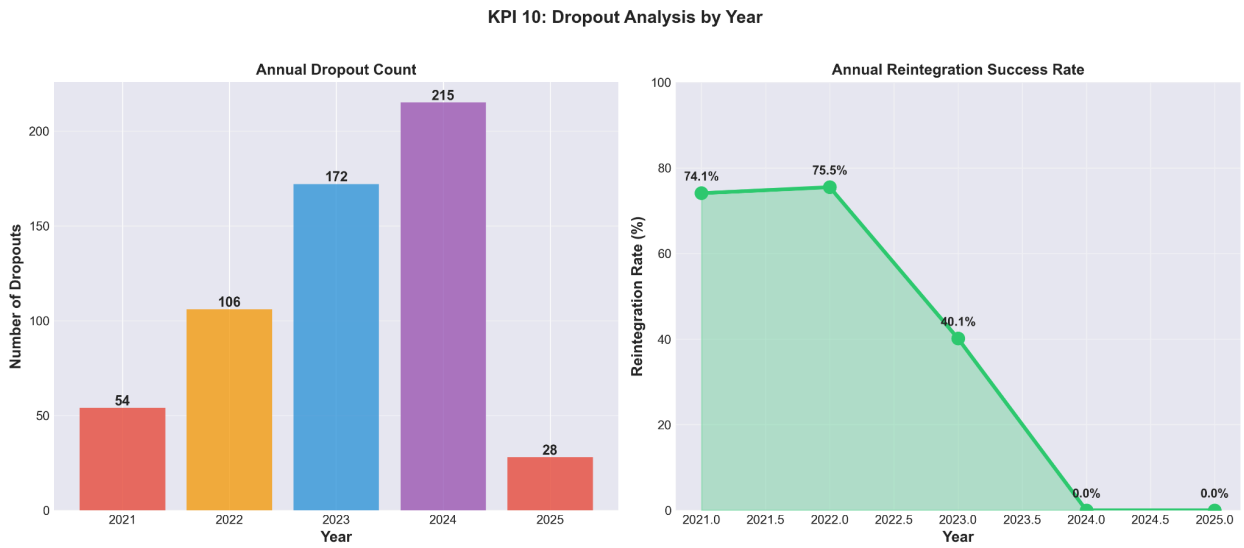
KPI 9: Dropout and Reintegration by Gender



This KPI compares overall dropout rates and reintegration outcomes between genders.

- Male students account for 290 dropouts with 111 reintegrations (38.3% success rate).
- Female students have 285 dropouts with only 78 reintegrations (27.4% success rate).
- While dropout numbers are nearly equal, males have a 40% higher reintegration success rate than females.
- This 11-percentage-point gap suggests gender-specific barriers prevent female students from returning to education.

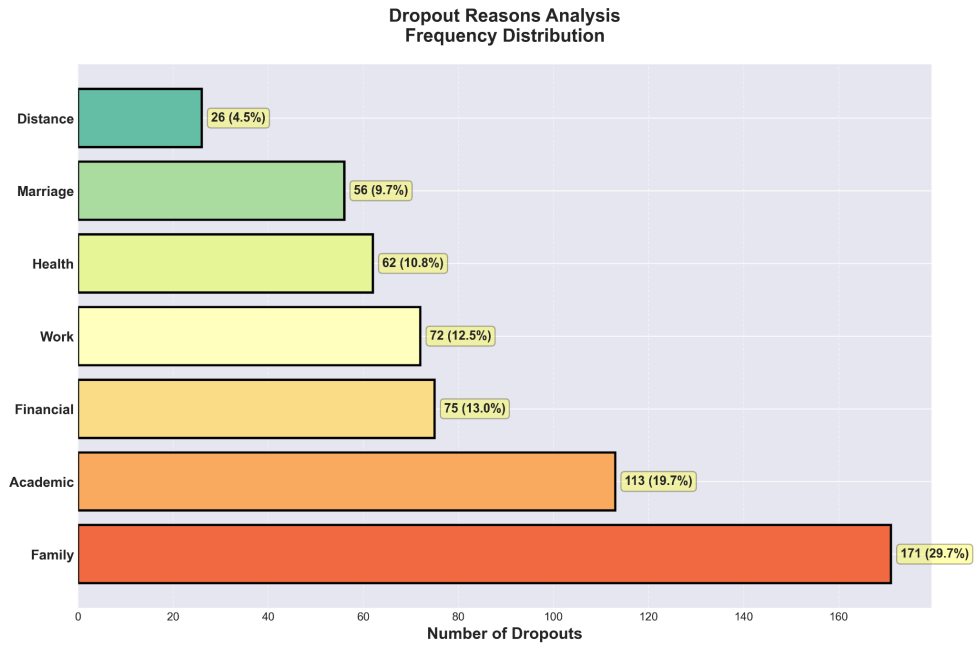
KPI 10: Dropout Analysis by Year



This KPI tracks annual trends in dropout counts and reintegration effectiveness.

- Dropout numbers increased dramatically from 54 in 2021 to a peak of 215 in 2024—a nearly 300% increase.
- 2025 shows only 28 dropouts, suggesting a major intervention or policy change.
- Reintegration success rates were strong in 2021 (74.1%) and 2022 (75.5%).
- Success rates collapsed to 40.1% in 2023, then dropped to 0% in both 2024 and 2025.
- The simultaneous decrease in dropouts and reintegration success by 2024-2025 may indicate that recent dropouts haven't had sufficient time for reintegration efforts, or that intervention programs have been discontinued.

KPI 11: Dropout Reasons Analysis



This KPI ranks the primary causes of student dropout across all cases.

- Family issues dominate with 171 cases (29.7%), representing nearly 1 in 3 dropouts.
- Academic difficulties account for 113 cases (19.7%), the second-leading cause.
- Financial constraints affect 75 students (13.0%).
- Work obligations cause 72 dropouts (12.5%).
- Health issues impact 62 students (10.8%).
- Marriage accounts for 56 cases (9.7%).
- Distance from school is the least common reason with only 26 cases (4.5%).
- Non-academic factors (family, financial, work, marriage, health, distance) collectively represent over 80% of all dropouts, suggesting that educational interventions alone cannot solve the dropout crisis.

8. CONCLUSION

8.1 Project Review

This project focused on designing and implementing a **Data Warehouse and Business Intelligence system** to analyze educational dropout and reintegration in Tunisia. The main objective was to collect data from different sources, clean and transform it using an ETL process, store it in a structured warehouse, and finally analyze it through dashboards and KPIs.

The project covered data from **two regions (Tunis and Sousse)** over several years and aimed to provide clear insights about dropout causes, affected groups, and the effectiveness of reintegration programs. By combining **Python, SQL, PostgreSQL, and Power BI**, the project demonstrates how data can support decision-making in the education sector.

8.2 Key Achievements

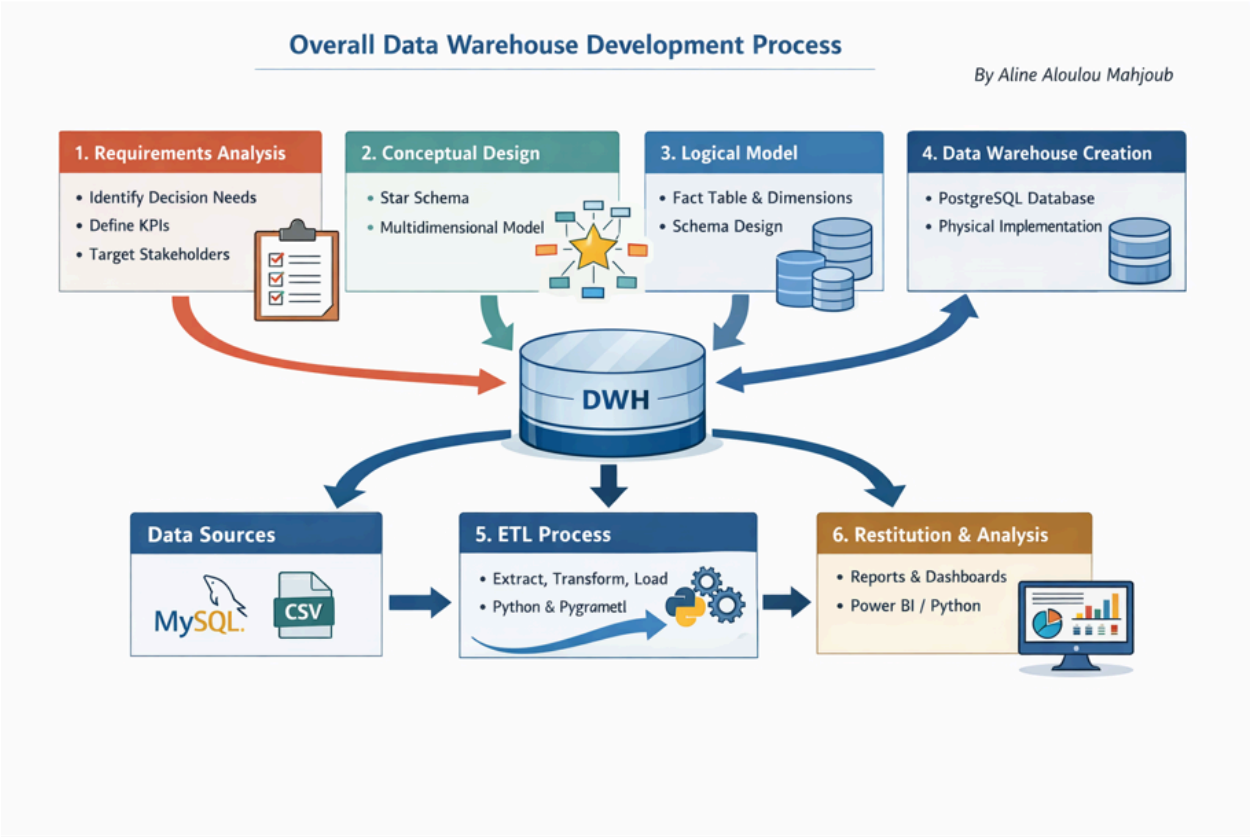
One of the main achievements of this project is the successful implementation of a **star schema** with multiple dimension tables and a fact table, which made analysis efficient and well organized.

Other important achievements include:

- Integration of **multiple data sources** (CSV files and relational databases)
- Implementation of a **complete ETL pipeline** with initial and incremental loading
- Use of **Slowly Changing Dimensions (Type 2)** to track student history over time
- Creation of **11 meaningful KPIs** answering real educational questions

- Development of **interactive Power BI dashboards** and static Python visualizations
- Identification of high-risk groups such as **middle school students, adolescents (12–18 years), and female students affected by early marriage**

These results show that the system is both technically correct and useful from a business and social perspective.



8.3 Difficulties Encountered

Several challenges were faced during the project:

- **Data quality problems:** The CSV files contained missing values, duplicates, and inconsistent formats. A large part of the work was dedicated to cleaning and validating the data.

- **Date format inconsistencies:** Dates appeared in multiple formats across sources, which required flexible parsing and careful validation.
- **Slowly Changing Dimensions:** Understanding and correctly implementing SCD Type 2 was challenging at first, especially when handling historical student records.
- **Incremental loading logic:** Avoiding duplicate records while still loading new data required the use of a control table and careful logic.

Despite these difficulties, each problem helped improve my understanding of real-world data engineering challenges.

8.4 Prospects for Improvement

This project can be improved and extended in several ways:

- Replace static dashboards with **fully interactive web dashboards** using tools like Streamlit or Dash
- Expand the data warehouse to include **all Tunisian regions** for national-level analysis
- Add **predictive models** to identify students at risk of dropping out
- Implement **automated alerts** to notify stakeholders when dropout rates increase

- Integrate data in real time through APIs or mobile data collection tools

These improvements would make the system more powerful and closer to a real production environment.

8.5 Final Reflections

This project was a very valuable learning experience. It showed that working with real data is very different from working with clean academic examples. Most of the effort was spent on **data cleaning, validation, and transformation**, which are critical steps in any BI project.

Beyond technical skills, this project helped me develop:

- Problem-solving abilities
- Better understanding of data-driven decision making
- Confidence in working with real datasets

Overall, this project demonstrates my ability to design a complete BI solution, from raw data to actionable insights, while addressing an important social issue through data analysis.